

Building the State in Real Time

The Ranking Problem: A Consultation · Applied Causal Inference, Week 7

August 31 – September 4, 2026 · Final memo due Sunday, September 6, 5:00pm

Harris Social Impact Fellowship · Summer 2026

Overview

This exercise is built directly from our research proposal to the President’s national security cabinet: the evaluation design you work on this week is the one the research team is developing with the Federal Cabinet of Public Security, and your output will flow into real policy-making. The Government of Mexico selected 61 priority municipalities for the ENSP intervention using four criteria: (i) intentional-homicide incidence in 2025; (ii) violent-robbery incidence in 2025; (iii) levels of or increases in perceived insecurity; (iv) strategic importance. The weighting scheme is not publicly disclosed. What is known, and what defines this week’s problem, is the **two-step structure of selection**: **53** municipalities entered by their ranking on criteria 1–3; the remaining **8** entered through criterion 4, strategic importance, off the ranking. The distributed treated list (`treated61.csv`) flags which municipalities entered by which route.

Your job mirrors the live research proposal’s Project 2: reconstruct the selection process well enough to support a credible evaluation design. That means solving both steps:

- **Step 1, the ranking**: reconstruct the latent index behind the 53, using *only* data on criteria 1–3. Every team works from the same criteria data; the candidate running variables are common property, not domain property
- **Step 2, the strategic additions**: characterize what put the other 8 in. The catalog provides twenty candidate “strategic importance” characteristics; the modeling is yours

The four teams remain distinct through their **domains**: economic indicators, demographic characteristics, health and human development, infrastructure. Domain variables do not enter any running variable. They are the **credibility material**: each team runs the smoothness battery at the induced threshold using its own domain’s pre-determined covariates, so the four presentations jointly deliver what no single team can: covariate smoothness across four independent slices of municipal life. Two deliverables: Friday, each team briefs the room (fifteen minutes, five of questions); by Sunday 5:00pm, each team submits a memo of at least five pages with a technical appendix, built alongside the deck and revised in light of Friday’s feedback.

Step 1: three candidate running variables (criteria 1–3 only)

From the shared criteria data (SESNP homicides 2025; SESNP violent robbery and its subcategories; ENSU/ENVIPE perceived insecurity, with its coverage limits documented):

- **Composite z-score**: standardize, orient, average (equal and inverse-variance weights)
- **Principal component score**: the first component; agnostic about what separates treated from untreated

- **Fisher discriminant (LDA) score:** the projection maximally separating the *53 rank-selected* municipalities from the never-treated. Train on the 53, not the 61: the 8 strategic additions are off-ranking by construction and would contaminate the discriminant. LDA uses the labels; report its fit as in-sample

Direction discipline throughout: higher score = higher security risk; rank 1 = worst. The pipeline enforces both with printed checks and hard assertions.

Validating Step 1 against the 53

- **Overlap at 53:** how many of your top-53 are among the 53 rank-selected?
- **Rank positions:** where do the 53 sit in your ranking; where do the 8 strategic municipalities sit? If the two-step story is right, the 8 should look *unranked*: scattered well below your rank-53 line
- **The induced first stage:** the share of your top-53 that is truly rank-selected is the treatment-probability jump your reconstruction delivers at the threshold; it is the first stage of the fuzzy design
- **Handling the 8 in the RD:** treated units sitting far above the cutoff on your running variable. State your choice and defend it: exclude them from the estimation sample, or keep them and absorb them into the fuzzy first stage. This is a real design decision the research team faces

Step 2: the strategic eight

Using the catalog’s twenty strategic-importance characteristics (border and port geography, trafficking corridors, high-profile violence, state presence, strategic economies):

- Compare the 8 against the municipalities that *could* have been strategic picks but were not: standardized differences on each characteristic
- Fit a simple classification (logit or LDA) of strategic selection among non-rank-selected municipalities. Eight positives is a small sample; report this as descriptive pattern-finding, never as a precise model. Which three or four characteristics carry the separation?
- Deliver the sentence the Cabinet evaluation needs: what does “strategic importance” appear to operationalize, and can the evaluation treat the 8 as ignorably assigned? (Anticipate: no. Then what?)

Credibility: the smoothness battery (your team’s domain-specific assignment)

At your preferred candidate’s rank-53 threshold, run the battery on **your domain’s** pre-determined covariates (≥ 20 processed and evaluated per team, per the catalog):

- **rdrobust** on each covariate, robust rows, read *jointly*: one marginal interval among twenty is noise; systematic jumps downweight the candidate
- **Density** on the constructed score (**rddensity**); ranks are uniform by construction and untestable

- **Placebo thresholds** and **donut** specifications, one flag each in the pipeline
- Covariates must be pre-determined: pre-2024 vintages, per the catalog's cross-section rules

An overview of this week's preliminary schedule

Daily rhythm

- **Team meetings with Austin**, Monday–Thursday mornings: twenty minutes per team in thirty-minute blocks from 10:00 (10:00, 10:30, 11:00, 11:30), the trailing ten minutes of each block open office hours. Order rotates daily; Monday runs Teams 1→4. Bring that day's by-noon deliverable in whatever state it is in
- **By-noon deliverables** are firm: submitted to Canvas by 12:00. **Stretch deliverables** are due by 5:00pm
- **Prabidhik is available 3:00–5:00pm only**: schedule code debugging and stretch work into that window

Monday, August 31: understanding context and data intake

- Read Project 2 of the proposal and the two-step selection structure; write the team's one-paragraph statement of the problem it is solving
- Portal orientation: claim your catalog variables and assign an owner to each within the team
- Build the key frame: read the Marco Geoestadístico shapefile, validate and reproject (EPSG:6372), produce the ~2,470-row CVEGEO frame with areas
- Assemble the criteria cross-section for criteria 1–2: SESNSP homicides and robbery subcategories collapsed to 2025 totals, per-100k rates with Censo denominators
- Run the pipeline key audit on everything assembled; draft the data-memo paragraph for the criteria table
- Sketch the perception plan for Tuesday: which ENSU urban areas map to which municipalities, and where coverage gaps will force an ENVIPE fallback

By noon: problem statement + key frame passing the audit + variable-ownership plan. **Stretch by 5pm:** criteria 1–2 cross-section passing the audit + its data-memo paragraph.

Tuesday, September 1: completing the criteria data and building the candidates

- Assemble perception (criterion 3): ENSU levels and changes where observed, ENVIPE state fallback elsewhere, coverage and imputation choices written down as made; merge into the criteria cross-section
- Fill the DIRECTION map for every criteria variable; run the printed sanity extremes; whole-team sign-off before anything downstream
- Build the composite candidate, equal-weight and inverse-variance versions
- Build the PCA candidate; verify the sign alignment against the composite
- Build the LDA candidate trained on the 53; verify in code that the 8 strategic municipalities are excluded from training
- Produce ranks; confirm the rank-1-equals-worst assertions pass; eyeball the top-53 and name the cases you expected to see

By noon: completed criteria cross-section with perception + direction-checked candidates. **Stretch by 5pm:** candidate table (overlap-at-53, induced first stage); rank-position plot of the 53 and the 8; ten or more domain covariates assembled.

Wednesday, September 2: validation and the strategic eight

- Complete Step-1 validation: rank positions of the 53 and the 8, separation plot, AUC; investigate the worst-ranked stragglers by name
- Choose the preferred candidate and write the three-sentence justification; compute its induced first stage, and if below ~ 0.5 , the weak-reconstruction paragraph
- Assemble the strategic-characteristics table; compute standardized differences between the 8 and the not-chosen pool
- Fit the descriptive logit, three regressors at most; draft the sentence stating what “strategic importance” appears to operationalize
- Complete the domain covariate set (≥ 20 , pre-2024 vintages), collapse rules recorded
- First pass of the smoothness battery on your domain covariates at the preferred threshold

By noon: validation figure array + the preferred-candidate justification. **Stretch by 5pm:** the Step-2 comparison complete; domain covariate set complete; first battery pass run.

Thursday, September 3: credibility and assembly

- Complete the battery on all ≥ 20 domain covariates; write the joint read
- Density test on the constructed score; placebo thresholds; donut flags
- Make the handling-the-8 decision: run the battery both ways (`KEEP_STRATEGIC` on and off), choose, write the defense
- Bandwidth sensitivity at the threshold; note anything that moves
- Build the deck to the Friday beat sheet; figure-quality pass (labeled, grayscale-readable)
- Dry-run the fifteen minutes; assign speakers and the Q&A lead

By noon: full battery + the handling-the-8 defense. **Stretch by 5pm:** draft deck submitted for overnight comments.

Friday, September 4: the briefing

Fifteen minutes, five of questions. Required beats: the criteria data and your candidates (3 min); validation against the 53 and the induced first stage (4); the strategic eight: what drives them, and your handling decision (3); your domain’s smoothness battery (3); what the reconstruction can and cannot support (2). Closing synthesis for the room: pooling all four domains’ batteries, does the rank-53 threshold hold up, and what first stage should the proposal’s fuzzy design expect?

Sunday, September 6, 5:00pm: the final memo

Built alongside the deck, submitted after Friday's feedback. Minimum five pages plus a technical appendix.

- **Body** (five pages minimum): stitches together every deck element in memo form: the problem and the two-step structure; the criteria data; the three candidates and validation against the 53; the induced first stage; the strategic eight and the handling decision; your domain's smoothness battery; what the reconstruction can and cannot support. Written for the research team briefing the Cabinet
- **Response to Friday:** a short section stating what the room's questions changed, and what they did not, and why
- **Technical appendix:** the full variable list with sources and processing rules; the complete candidate results and battery tables; density, placebo, donut, and bandwidth outputs; the annotated code the results were produced from

The memo is the deliverable that travels: its findings feed the research team's own reconstruction.